Other permitted identification or labelling systems

In specific cases, a WHMIS label can also be a mark, sign, stamp, sticker, seal, ticket, tag, or wrapper. It can be attached, imprinted, stencilled, or embossed on the hazardous product or its container. Workers must be trained to be able to identify these alternate systems if they are used in the workplace.

Variations on supplier and workplace labels apply for specific items such as:

- **Bulk shipments** — A labelling exemption exists for products sold without packaging.
- **Small capacity containers 100 mL or less** — Exempt only from the requirement to have precautionary or hazard statements on the label.
- **Small capacity containers 3 mL or less** — Where the label will interfere with normal use of the product, the product would be required to have a label that is durable and legible for transport and storage, but removable during use.
- **Piping systems and vessels** — When hazardous products are contained or transferred in a piping system, a process or reaction vessel, or a tank car, tank truck, ore car, conveyor belt, or similar conveyance, the use of labels, placards, colour coding, or any other mode of identification is allowed (along with worker training).



Hazardous products in pipes identified by colours and letters

Labelling laboratory samples

The Hazardous Products Regulations define a laboratory sample as a hazardous product packaged in a container that holds less than 10 kg of the product and is intended only to be tested in a laboratory. This definition excludes samples that will be used for testing other products or for educational or demonstration purposes.

Many laboratory samples are sent or received in small capacity containers of 100 mL or less. For this size of laboratory sample, the container's label does not have to include precautionary or hazard statements. However, the label must show the product identifier, pictogram(s), signal word, and supplier identifier. If the container size is 3 mL or less and the label would interfere with normal use, the label may be removed while working with the hazardous product.

Laboratory samples of biohazardous infectious materials have been given special exceptions for labels. Modified supplier labels are permitted. When a sample classified as "Biohazardous infectious materials (Category 1)" is sent to a laboratory, the attached label must show the following:

- The name of any hazardous chemical in the sample
- The name of the biohazardous infectious material (if known)
- The following statement:

  "Hazardous Laboratory Sample. For hazard information or in an emergency, call / Échantillon pour laboratoire de produit dangereux. Pour obtenir des renseignements sur les dangers ou en cas d'urgence, composez"

  This statement must be followed by an emergency telephone number.

In addition, labelling exceptions exist for laboratory samples of biohazardous infectious materials that are:

- Transferred to different containers than the ones received by the laboratory
- Collected or produced in the workplace

The exceptions allow for these hazardous products to be clearly identified through a combination of:

- A labelling system (for example, identifier labels, bar codes, or colour banding), and
- Worker education and training

For more information about exemptions for labelling laboratory samples and other containers, see Part 5 of the Hazardous Products Regulations.

What workers should do when using a hazardous product

As a worker, you should do the following:

- Always check to see if there is a label on the product before you use it.
- Read, understand, and follow the instructions on the label and SDS. And follow any education, instructions, and training your employer provides.
- Ask your supervisor if you are not sure about how to use or store the product.
- Ask for a new label when you can't see or read the old one properly.
- Don't use a product that is not labelled or if the label is unreadable. Ask your supervisor for help (for example, to replace the label).

5 Safety data sheets

About safety data sheets

What's new

- MSDS replaced by SDS.
- Standard 16-section format.
- New information requirements (for example, inclusion of the WHMIS classification, hazard statements, and other label elements in Section 2).
- SDSs must be accurate at the time of sale or import, for each sale or import.
- SDSs to be updated when significant new information becomes available.

Safety data sheets (SDSs) are documents that provide information about hazardous products and advice about safety precautions.

An SDS tells you what the hazards of a product are, how to use the product safely, what to expect if the recommendations are not followed, how to recognize symptoms of exposure, and what to do if emergencies occur.

SDSs provide more-detailed hazard information about products than labels do. SDSs are important resources that help you learn more about the products you use. Use this information to identify:

- The hazards of the products you use
- How to protect yourself from those hazards
- Safe handling and emergency measures

SDSs are usually prepared or obtained by the manufacturer or importer of a product. In some circumstances, an employer may be required to prepare an SDS (for example, when the product is produced and used exclusively in that workplace).

Every product that is classified as a hazardous product under WHMIS that is intended for use, handling, or storage in a workplace in Canada must have an SDS.

Format and information required

WHMIS 2015 requires a standard 16-section SDS. All information on the SDS must appear in the specified order.

The following table gives an overview of the information to be provided in each section. (A full list of the required information appears on page 79.)

Table 7. Overview of information requirements for each section of an SDS

SDS section and heading	Information requirements (partial list)
1 Identification	Product identifier, recommended use and restrictions on use, supplier contact information, emergency phone number.
2 Hazard identification	Classification (hazard class and category), label elements (including pictogram, signal word, hazard statements, and precautionary statements), and other hazards (e.g., thermal hazards).
3 Composition/ Information on ingredients	For a hazardous product that is a substance: the chemical name, synonyms, Chemical Abstracts Service Number (CAS No.), and the chemical name of impurities, stabilizing solvents, and stabilizing additives where classified and that contribute to the classification of the product. For a hazardous product that is a mixture: for ingredients that present a health hazard, the chemical name, synonyms, CAS No., and concentration. Note: Confidential business information rules may apply (see page 67 for more information).
4 First-aid measures	First-aid measures by route of exposure (inhalation, skin contact, etc.) as well as most important symptoms/effects.
5 Fire-fighting measures	Suitable (and unsuitable) extinguishing media, specific hazards, special equipment and precautions for firefighters.
6 Accidental release measures	Protective equipment, emergency procedures, methods and materials for containment and cleanup.
7 Handling and storage	Precautions for safe handling, conditions for storage, including any incompatibilities.
8 Exposure controls/ Personal protection	Exposure limits, engineering controls, personal protective equipment.
9 Physical and chemical properties	Appearance, odour, odour threshold, pH, melting/freezing point, boiling point and range, flash point, upper and lower flammable or explosive limits.
10 Stability and reactivity	Reactivity, chemical stability, possible hazardous reactions, conditions to avoid, incompatible materials, hazardous decomposition products.

SDS section and heading	Information requirements (partial list)
11 Toxicological information	Description of various toxic effects by route of entry, including effects of acute or chronic exposure, carcinogenicity, reproductive effects, respiratory sensitization.
12 Ecological information*	Aquatic and terrestrial toxicity (if available), persistence and degradability, bioaccumulative potential, mobility in soil.
13 Disposal considerations*	Safe handling and methods of disposal, including contaminated packaging.
14 Transport information*	UN number and proper shipping name, hazard classes, packing group.
15 Regulatory information*	Safety, health, and environmental regulations specific to the product.
16 Other information	Other information, including date of the latest revision of the SDS.

* Sections 12 to 15 require the headings to be present. The supplier has the option to not provide information in these sections.

Why SDSs can be difficult to understand

SDSs are complex and technical. SDSs have many different audiences, including occupational hygienists and safety professionals, employers, supervisors, nurses, doctors, emergency responders, and workers.

To ensure that SDS users can quickly find the information they need, information directed toward these various users will be listed in specific sections. Having a set format will make it easier to find the information you need on every SDS.

However, workers may find some of the information on an SDS difficult to understand. Employers must be able to explain the content of the SDS to workers in order for them to work safely with or near hazardous products.

The following sample SDS shows information items for acetone.

Rappter Chemicals

Acetone

Safety Data Sheet

Date of issue: November 12, 1998 Revision date: June 1, 2018 Supersedes: May 10, 2015 Version: 1.3

SECTION 1: Identification

Substance name	Acetone
CAS No	67-64-1
Product code	BC987458
Formula	C3H6O
Synonyms	2-propanone / beta-ketopropane / dimethyl formaldehyde / dimethyl ketone / dimethylketal / DMK (=dimethyl ketone) / keto propane / methyl ketone / pyroacetic acid / pyroacetic ether / pyroacetic spirit

1.1. Recommended uses

Use of the substance : Solvent
Cleansing product Chemical raw material

1.2. Supplier Contact Information

Rappter Chemicals Ltd.
1234 Jurassic Court
Moose Jaw, SK
T 123-456-7890 - F 987-654-3210

1.3. Emergency Telephone Number

Emergency number: 123-456-7890 ext 4357 : CHEMTREC: 1-800-424-9300 or 011-703-527-3887

SECTION 2: Hazard identification

2.1. Classification

WHMIS 2015 classification

Flammable Liquid 2;

Eye Irritation 2A;

Specific Target Organ Toxicity (STOT), Single Exposure (SE), Narcotic Effects 3

2.2. Label elements

WHMIS 2015 labeling

Hazard pictograms (WHMIS 2015)



GHS02 GHS07

Signal word (WHMIS 2015) : Danger

Hazard statements (WHMIS 2015) : H225 - Highly flammable liquid and vapour H319 - Causes serious eye irritation
H336 - May cause drowsiness or dizziness

Precautionary statements (WHMIS 2015) : P210 - Keep away from heat, hot surfaces, open flames, sparks, and other ignition sources. - No smoking P233 - Keep container tightly closed
P240 - Ground and bond container and receiving equipment
P241 - Use explosion-proof electrical, lighting, ventilating equipment P242 - Use only non-sparking tools
P243 - Take action to prevent static discharges P261 - Avoid breathing mist, spray, vapours
P264 - Wash exposed skin thoroughly after handling P271 - Use only outdoors or in a well-ventilated area
P280 - Wear eye protection, face protection, protective clothing, protective gloves
P303+P361+P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower
P304+P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing
P305+P351+P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
P312 - Call a POISON CENTER if you feel unwell P337+P313 - If eye irritation persists: Get medical attention
P370+P378 - In case of fire: Use dry chemical powder, alcohol-resistant foam, carbon dioxide (CO2) for extinction
P403+P233 - Store in a well-ventilated place. Keep container tightly closed

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P405 - Store locked up
P501 - Dispose of container to comply with local, state and federal regulations P235 - Keep cool

2.3. Other hazards

Other hazards not contributing to the classification : None

2.4. Unknown acute toxicity (WHMIS 2015)

No data available

SECTION 3: Composition/Information on ingredients

3.1. Substances

Name	Product identifier	%	WHMIS 2015 classification
Acetone (Main constituent)	(CAS No) 67-64-1	100	Flam. Liq. 2, H225 Eye Irrit. 2A, H319 STOT SE 3, H336

Full text of H-phrases: see section 16

3.2. Mixture

Not applicable

SECTION 4: First-aid measures

4.1. Description of first aid measures

First-aid measures general : Check the vital functions. Unconscious: maintain adequate airway and respiration. Respiratory arrest: artificial respiration or oxygen. Cardiac arrest: perform resuscitation. Victim conscious with laboured breathing: half-seated. Victim in shock: on his back with legs slightly raised. Vomiting: prevent asphyxia/aspiration pneumonia. Prevent cooling by covering the victim (no warming up). Keep watching the victim. Give psychological aid. Keep the victim calm, avoid physical strain. Depending on the victim's condition: doctor/hospital.

First-aid measures after inhalation : Remove the victim into fresh air. Respiratory problems: consult a doctor/medical service.

First-aid measures after skin contact : Wash immediately with lots of water. Soap may be used. Do not apply (chemical) neutralizing agents. Remove clothing before washing. Take victim to a doctor if irritation persists.

First-aid measures after eye contact : Rinse immediately with plenty of water. Do not apply neutralizing agents. Take victim to an ophthalmologist if irritation persists.

First-aid measures after ingestion : Rinse mouth with water. Immediately after ingestion: give lots of water to drink. Do not give milk/oil to drink. Do not induce vomiting. Give activated charcoal. Call Poison Information Centre (www.big.be/antigif.htm). Consult a doctor/medical service if you feel unwell. Ingestion of large quantities: immediately to hospital. Doctor: gastric lavage.

4.2. Most important symptoms and effects, both acute and delayed

Symptoms/injuries : Not expected to present a significant hazard under anticipated conditions of normal use.

Symptoms/injuries after inhalation : EXPOSURE TO HIGH CONCENTRATIONS: Feeling of weakness. Irritation of the respiratory tract. Nausea. Vomiting. Headache. Central nervous system depression. Dizziness. Narcosis. Excited/restless. Drunkenness. Disturbed motor response. Respiratory difficulties. Disturbances of consciousness.

Symptoms/injuries after skin contact : ON CONTINUOUS EXPOSURE/CONTACT: Dry skin. Cracking of the skin.

Symptoms/injuries after eye contact : Irritation of the eye tissue.

Symptoms/injuries after ingestion : Dry/sore throat. Risk of aspiration pneumonia. Symptoms similar to those listed under inhalation. AFTER ABSORPTION OF HIGH QUANTITIES: Irritation of the gastric/intestinal mucosa. Change in the haemogramme/blood composition. Change in urine output. Affection of the renal tissue. Enlargement/affection of the liver.

Symptoms/injuries upon intravenous administration : Not available.

Chronic symptoms : ON CONTINUOUS/REPEATED EXPOSURE/CONTACT: Red skin. Skin rash/inflammation. Dry/sore throat. Headache. Nausea. Feeling of weakness. Loss of weight. Possible inflammation of the respiratory tract.

4.3. Indication of any immediate medical attention and special treatment needed

Obtain medical assistance.

SECTION 5: Fire-fighting measures

5.1. Extinguishing media

Suitable extinguishing media : Preferably: alcohol resistant foam. Water spray. Polyvalent foam. BC powder. Carbon dioxide.

Unsuitable extinguishing media : Solid water jet ineffective as extinguishing medium.

5.2. Special hazards arising from the substance or mixture

Fire hazard : DIRECT FIRE HAZARD. Highly flammable. Gas/vapour flammable with air within explosion limits. INDIRECT FIRE HAZARD. May be ignited by sparks. Gas/vapour spreads at floor level: ignition hazard. Reactions involving a fire hazard: see "Reactivity Hazard".

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Explosion hazard	:	DIRECT EXPLOSION HAZARD. Gas/vapour explosive with air within explosion limits. INDIRECT EXPLOSION HAZARD. Heat may cause pressure rise in tanks/drums: explosion risk. may be ignited by sparks. Reactions with explosion hazards: see "Reactivity Hazard".
Reactivity	:	Upon combustion: CO and CO2 are formed. Violent to explosive reaction with many compounds. Prolonged storage: on exposure to light: release of harmful gases/vapours. Reacts violently with (strong) oxidizers: peroxidation resulting in increased fire or explosion risk.

5.3. Advice for firefighters

Firefighting instructions	:	Cool tanks/drums with water spray/remove them into safety. Physical explosion risk: extinguish/cool from behind cover. Do not move the load if exposed to heat. After cooling: persistent risk of physical explosion.
Protection during firefighting	:	Heat/fire exposure: compressed air/oxygen apparatus.

SECTION 6: Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

6.1.1. For non-emergency personnel

Protective equipment	:	Gloves. Protective goggles. Protective clothing. Large spills/in enclosed spaces: compressed air apparatus.
Emergency procedures	:	Keep upwind. Mark the danger area. Consider evacuation. Seal off low-lying areas. Close doors and windows of adjacent premises. Stop engines and no smoking. No naked flames or sparks. Spark- and explosionproof appliances and lighting equipment. Keep containers closed. Wash contaminated clothes.

6.1.2. For emergency responders

Protective equipment	:	Equip cleanup crew with proper protection.
Emergency procedures	:	Ventilate area.

6.2. Environmental precautions

Prevent spreading in sewers.

6.3. Methods and material for containment and cleaning up

For containment	:	Contain released substance, pump into suitable containers. Consult "Material-handling" to select material of containers. Plug the leak, cut off the supply. Dam up the liquid spill. Try to reduce evaporation. Measure the concentration of the explosive gas-air mixture. Dilute/disperse combustible gas/vapour with water curtain. Provide equipment/receptacles with earthing. Do not use compressed air for pumping over spills.
Methods for cleaning up	:	Take up liquid spill into inert absorbent material, e.g.: sand, earth, vermiculite. Scoop absorbed substance into closing containers. See "Material-handling" for suitable container materials. Spill must not return in its original container. Carefully collect the spill/leftovers. Damaged/cooled tanks must be emptied. Do not use compressed air for pumping over spills. Clean contaminated surfaces with an excess of water. Take collected spill to manufacturer/competent authority. Wash clothing and equipment after handling.

6.4. Reference to other sections

See Heading 8. Exposure controls and personal protection.

SECTION 7: Handling and storage

7.1. Precautions for safe handling

Precautions for safe handling	:	Comply with the legal requirements. Remove contaminated clothing immediately. Clean contaminated clothing. Handle uncleaned empty containers as full ones. Thoroughly clean/dry the installation before use. Do not discharge the waste into the drain. Do not use compressed air for pumping over. Use spark-/explosionproof appliances and lighting system. Take precautions against electrostatic charges. Keep away from naked flames/heat. Keep away from ignition sources/sparks. Avoid prolonged and repeated contact with skin. Keep container tightly closed. Measure the concentration in the air regularly. Work under local exhaust/ventilation.
Hygiene measures	:	Do not eat, drink or smoke when using this product. Wash contaminated clothing before reuse. Wash hands and other exposed areas with mild soap and water before eating, drinking or smoking and when leaving work.

7.2. Conditions for safe storage, including any incompatibilities

Storage conditions	:	Keep only in the original container in a cool, well ventilated place away from : Heat sources, Direct sunlight, incompatible materials. Keep container closed when not in use.
Incompatible products	:	Strong bases. Strong acids.
Incompatible materials	:	Sources of ignition. Direct sunlight.
Storage temperature	:	15 - 20 °C
Heat and ignition sources	:	KEEP SUBSTANCE AWAY FROM: heat sources. ignition sources.
Prohibitions on mixed storage	:	KEEP SUBSTANCE AWAY FROM: oxidizing agents. reducing agents. (strong) acids. (strong) bases. halogens. amines.

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Storage area	:	Store in a cool area. Keep out of direct sunlight. Store in a dry area. Store in a dark area. Ventilation at floor level. Fireproof storeroom. Provide for an automatic sprinkler system. Provide for a tub to collect spills. Provide the tank with earthing. Meet the legal requirements.
Special rules on packaging	:	SPECIAL REQUIREMENTS: closing. with pressure relief valve. clean. opaque. correctly labelled. meet the legal requirements. Secure fragile packagings in solid containers.
Packaging materials	:	SUITABLE MATERIAL: steel. stainless steel. carbon steel. aluminium. iron. copper. nickel. bronze. glass. MATERIAL TO AVOID: synthetic material.

7.3. Specific end use(s)

No additional information available

SECTION 8: Exposure controls/Personal protection

8.1. Control parameters

Acetone (67-64-1)		
BC OHS Regulation	TWA (ppm)	250 ppm
BC OHS Regulation	STEL/Ceiling (ppm)	500 ppm
ACGIH	TWA (ppm)	500 ppm
ACGIH	STEL (ppm)	750 ppm
OSHA	PEL (TWA) (mg/m³)	2400 mg/m³
OSHA	PEL (TWA) (ppm)	1000 ppm

8.2. Exposure controls

Appropriate engineering controls	:	Emergency eye wash fountains and safety showers should be available in the immediate vicinity of any potential exposure.
Materials for protective clothing	:	GIVES EXCELLENT RESISTANCE: No data available. GIVES GOOD RESISTANCE: butyl rubber. tetrafluoroethylene. GIVES LESS RESISTANCE: chlorosulfonated polyethylene. natural rubber. neoprene. polyurethane. PVA. styrene-butadiene rubber. GIVES POOR RESISTANCE: nitrile rubber. polyethylene. PVC. viton. nitrile rubber/PVC.
Hand protection	:	Gloves (see information above).
Eye protection	:	Protective goggles.
Skin and body protection	:	Head/neck protection. Protective clothing.
Respiratory protection	:	Wear half mask respirator with Organic Vapour cartridges as a minimum.
Other information	:	Do not eat, drink or smoke during use.

SECTION 9: Physical and Chemical Properties

Information on basic physical and chemical properties

Physical state	Liquid
Appearance	: Liquid.
Molecular mass	: 58.08 g/mol
Colour	: Colourless.
Odour	: Aromatic odour. Sweet odour. Fruity odour.
Odour threshold	: 306 - 653 ppm 737 - 1574 mg/m³
pH	: 7
Relative evaporation rate (butylacetate=1)	: 6
Relative evaporation rate (ether=1)	: 2
Melting point	: -95 °C
Freezing point	: No data available
Boiling point	: 56 °C
Flash point	: -18 °C
Critical temperature	: 235 °C
Self ignition temperature	: 465 °C
Decomposition temperature	: No data available
Flammability (solid, gas)	: No data available
Vapour pressure	: 247 hPa
Vapour pressure at 50 °C	: 828 hPa
Critical pressure	: 47010 hPa
Relative vapour density at 20 °C	: 2.0
Relative density	: 0.79
Relative density of saturated gas/air mixture	: 1.2

Density	:	786 kg/m³
Solubility	:	Soluble in water. Soluble in ethanol. Soluble in ether. Soluble in dimethyl ether. Soluble in petroleum spirit. Soluble in chloroform. Soluble in dimethylformamide. Soluble in oils/fats. Water: Complete Ethanol: Complete Ether: Complete
Log Pow	:	-0.24 (Test data)
Log Kow	:	No data available
Viscosity, kinematic	:	0.417 mm²/s
Viscosity, dynamic	:	0.00033 Pa.s
Explosive properties	:	No data available
Oxidizing properties	:	None
Explosive limits	:	2 - 12.8 vol % 60 - 310 g/m³

9.1. Other information

Minimum ignition energy		1.15 mJ
Specific conductivity	:	500000 pS/m
Saturation concentration	:	589 g/m³
VOC content	:	100 %
Other properties	:	Gas/vapour heavier than air at 20°C. Clear. Highly volatile. Substance has neutral reaction.

SECTION 10: Stability and reactivity

10.1. Reactivity

Upon combustion: CO and CO2 are formed. Violent to explosive reaction with many compounds. Prolonged storage: on exposure to light: release of harmful gases/vapours. Reacts violently with (strong) oxidizers: peroxidation resulting in increased fire or explosion risk.

10.2. Chemical stability

Unstable on exposure to light.

10.3. Possibility of hazardous reactions

Not established.

10.4. Conditions to avoid

Direct sunlight. Extremely high or low temperatures.

10.5. Incompatible materials

Strong acids. Strong bases.

10.6. Hazardous decomposition products

Carbon monoxide. Carbon dioxide.

SECTION 11: Toxicological information

11.1. Information on toxicological effects

Acute toxicity : Not classified

Acetone (\f)67-64-1	
LD50 oral rat	5800 mg/kg (Rat; Experimental value,Rat; Experimental value)
LD50 dermal rabbit	20000 mg/kg (Rabbit; Experimental value,Rabbit; Experimental value)
LC50 inhalation rat (mg/l)	71 mg/l/4h (76 mg/l/4h; Rat; Rat; Experimental value; Experimental value,76 mg/l/4h; Rat; Rat; Experimental value; Experimental value)
LC50 inhalation rat (ppm)	30000 ppm/4h (Rat; Experimental value,Rat; Experimental value)

Skin corrosion/irritation : Not classified pH: 7

Serious eye damage/irritation : Causes serious eye irritation. pH: 7

Respiratory or skin sensitisation : Not classified

Germ cell mutagenicity : Not classified
Based on available data, the classification criteria are not met

Carcinogenicity : Not classified

Reproductive toxicity : Not classified
Based on available data, the classification criteria are not met

Specific target organ toxicity (single exposure) : May cause drowsiness or dizziness.

Specific target organ toxicity (repeated exposure)	:	Not classified Based on available data, the classification criteria are not met
Aspiration hazard	:	Not classified Based on available data, the classification criteria are not met
Potential Adverse human health effects and symptoms	:	Based on available data, the classification criteria are not met.
Symptoms/injuries after inhalation	:	EXPOSURE TO HIGH CONCENTRATIONS: Feeling of weakness. Irritation of the respiratory tract. Nausea. Vomiting. Headache. Central nervous system depression. Dizziness. Narcosis. Excited/restless. Drunkenness. Disturbed motor response. Respiratory difficulties. Disturbances of consciousness.
Symptoms/injuries after skin contact	:	ON CONTINUOUS EXPOSURE/CONTACT: Dry skin. Cracking of the skin.
Symptoms/injuries after eye contact	:	Irritation of the eye tissue.
Symptoms/injuries after ingestion	:	Dry/sore throat. Risk of aspiration pneumonia. Symptoms similar to those listed under inhalation. AFTER ABSORPTION OF HIGH QUANTITIES: Irritation of the gastric/intestinal mucosa. Change in the haemogramme/blood composition. Change in urine output. Affection of the renal tissue. Enlargement/affection of the liver.
Symptoms/injuries upon intravenous administration	:	Not available.
Chronic symptoms	:	ON CONTINUOUS/REPEATED EXPOSURE/CONTACT: Red skin. Skin rash/inflammation. Dry/sore throat. Headache. Nausea. Feeling of weakness. Loss of weight. Possible inflammation of the respiratory tract.

SECTION 12: Ecological information

12.1. Toxicity

Ecology - general : Classification concerning the environment: not applicable.

Ecology - air : TA-Luft Klasse 5.2.5.

Ecology - water : Not harmful to fishes (LC50(96h) >1000 mg/l). Not harmful to invertebrates (Daphnia). Not harmful to algae (EC50 >1000 mg/l). Not harmful to plankton. Inhibition of activated sludge.

Acetone (67-64-1)	
LC50 fishes 1	6210 mg/l (96 h; Pimephales promelas; NOMINAL CONCENTRATION)
EC50 Daphnia 1	8800 mg/l (48 h; Daphnia pulex)
LC50 fish 2	5540 mg/l 96 h; Salmo gairdneri (Oncorhynchus mykiss)
TLM fish 1	13000 ppm (96 h; Gambusia affinis; TURBULENT WATER)
TLM fish 2	> 1000 ppm (96 h; Pisces)
Threshold limit other aquatic organisms 1	3000 mg/l (Plankton)
Threshold limit other aquatic organisms 2	28 mg/l (Protozoa)
Threshold limit algae 1	7500 mg/l (Scenedesmus quadricauda; PH = 7)
Threshold limit algae 2	3400 mg/l (48 h; Chlorella sp.)

12.2. Persistence and degradability

Acetone (67-64-1)	
Persistence and degradability	Readily biodegradable in water. Biodegradable in the soil. Biodegradable in the soil under anaerobic conditions. No (test)data on mobility of the substance available.
Biochemical oxygen demand (BOD)	1.43 g O^2/g substance
Chemical oxygen demand (COD)	1.92 g O^2/g substance
ThOD	2.20 g O^2/g substance

12.3. Bioaccumulative potential

Acetone (67-64-1)	
BCF fish 1	0.69 (Pisces)
BCF other aquatic organisms 1	3
Log Pow	-0.24 (Test data)
Bioaccumulative potential	Not bioaccumulative.

12.4. Mobility in soil

Acetone (67-64-1)	
Surface tension	0.0237 N/m

12.5. Other adverse effects

Other information : Avoid release to the environment.

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SECTION 13: Disposal considerations

13.1. Waste treatment methods

Waste disposal recommendations : Remove waste in accordance with local and/or national regulations. Hazardous waste shall not be mixed together with other waste. Different types of hazardous waste shall not be mixed together if this may entail a risk of pollution or create problems for the further management of the waste. Hazardous waste shall be managed responsibly. All entities that store, transport or handle hazardous waste shall take the necessary measures to prevent risks of pollution or damage to people or animals. Recycle by distillation. Remove to an authorized waste incinerator for solvents with energy recovery. Do not discharge into drains or the environment.

Additional information : LWCA (the Netherlands): KGA category 03. Hazardous waste according to Directive 2008/98/EC.

Ecology - waste materials : Avoid release to the environment.

SECTION 14: Transportation Information

14.1. UN number

UN-No.(DOT) 1090

DOT NA no. UN1090

14.2. UN proper shipping name

DOT Proper Shipping Name : Acetone

Department of Transportation (DOT) Hazard Classes : 3 - Class 3 - Flammable and combustible liquid 49 CFR 173.120

Hazard labels (DOT) : 3 - Flammable liquids



Packing group (DOT) : II - Medium Danger

DOT Special Provisions (49 CFR 172.102) : IB2 - Authorized IBCs: Metal (31A, 31B and 31N); Rigid plastics (31H1 and 31H2); Composite (31HZ1). Additional Requirement: Only liquids with a vapor pressure less than or equal to 110 kPa at 50 C (1.1 bar at 122 F), or 130 kPa at 55 C (1.3 bar at 131 F) are authorized.
T4 - 2.65 178.274(d)(2) Normal..... 178.275(d)(3)
TP1 - The maximum degree of filling must not exceed the degree of filling determined by the following: Degree of filling = 97 / (1 + a (tr - tf)) Where: tr is the maximum mean bulk temperature during transport, and tf is the temperature in degrees celsius of the liquid during filling.

DOT Packaging Exceptions (49 CFR 173.xxx) : 150

DOT Packaging Non Bulk (49 CFR 173.xxx) : 202

DOT Packaging Bulk (49 CFR 173.xxx) : 242

14.3. Additional information

Other information : No supplementary information available.

State during transport (ADR-RID) : as liquid.

Overland transport

Packing group (ADR) : II

Class (ADR) : 3 - Flammable liquids

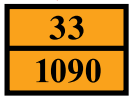
Hazard identification number (Kemler No.) : 33

Classification code (ADR) : F1

Danger labels (ADR) : 3 - Flammable liquids



Orange plates



Tunnel restriction code : D/E

Transport by sea

DOT Vessel Stowage Location : B - (i) The material may be stowed "on deck" or "under deck" on a cargo vessel and on a passenger vessel carrying a number of passengers limited to not more than the larger of 25 passengers, or one passenger per each 3 m of overall vessel length; and (ii) "On deck only" on passenger vessels in which the number of passengers specified in paragraph (k)(2)(i) of this section is exceeded.

EmS-No. (1) : F-E

EmS-No. (2) : S-D

Air transport

DOT Quantity Limitations Passenger aircraft/rail (49 CFR 173.27) : 5 L

DOT Quantity Limitations Cargo aircraft only (49 CFR 175.75) : 60 L

SECTION 15: Regulatory information

15.1. US Federal regulations

Acetone (67-64-1)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory	
RQ (Reportable quantity, section 304 of EPA's List of Lists) :	5000 lb

EU-Regulations

No additional information available

Classification according to Regulation (EC) No. 1272/2008 [CLP]

Flam. Liq. 2 H225 Eye Irrit. 2 H319 STOT SE 3 H336

Full text of H-phrases: see section 16

Classification according to Directive 67/548/EEC or 1999/45/EC

F; R11 Xi; R36 R66 R67

Full text of R-phrases: see section 16

15.2. US State regulations

No additional information available

SECTION 16: Other information

Other information : Information provided is current as of the latest revision date: June 1, 2018.

Full text of H-phrases:

Eye Irrit. 2A	Serious eye damage/eye irritation, Category 2A
Flam. Liq. 2	Flammable liquids, Category 2
STOT SE 3	Specific target organ toxicity — Single exposure, Category 3, Narcosis
H225	Highly flammable liquid and vapour
H319	Causes serious eye irritation
H336	May cause drowsiness or dizziness



NFPA health hazard : 1 - Exposure could cause irritation but only minor residual injury even if no treatment is given.

NFPA fire hazard : 3 - Liquids and solids that can be ignited under almost all ambient conditions.

NFPA reactivity : 0 - Normally stable, even under fire exposure conditions, and are not reactive with water.

Rappter Chemicals

Acetone
Safety Data Sheet

HMIS III Rating

Health	: 1 Slight Hazard - Irritation or minor reversible injury possible
Flammability	: 3 Serious Hazard
Physical	: 0 Minimal Hazard
Personal Protection	: C

SDS US (GHS HazCom 2012)

Information in this SDS is from available published sources and is believed to be accurate. No warranty, express or implied, is made and Rappter Chemicals LTD assumes no liability resulting from the use of this SDS. The user must determine suitability of this information for his application.

When workers should use SDSs

As a worker, you should keep the following points in mind.

Always be familiar with the hazards of a product **before** you start using it.

- Look at an SDS and match the name of the product on the container to the one on the SDS (Section 1).
- Know the hazards (Section 2).
- Understand safe handling and storage instructions (Section 7).
- Understand what to do in an emergency (sections 4, 5, and 6).

You can think of the SDS as having four main purposes. It provides information on:

- **Identification** — for the product and supplier
- **Hazards** — physical (fire and reactivity) and health
- **Prevention** — steps you can take to work safely and reduce or prevent exposure
- **Response** — appropriate responses in various emergency situations (for example, first aid, fire, or accidental release)

A few things to know:

- Make sure that the product is being used in the way the manufacturer intended. Otherwise, the advice provided on the SDS and label may not apply, or the protective measures listed may not be adequate. Section 1 of the SDS should describe the typical use of the product and may indicate restrictions. Ask your supervisor or a health and safety professional for advice if the way you use the product does not match the SDS.
- Section 2 will summarize the hazards related to the product, precautions to take, and what to do in an emergency. The SDS covers information about the potential hazards, but it may not be specific about the required safe work procedures needed for your workplace. (For example, the SDS may not specify what type of respirator must be used, just that a respirator is needed.) Ask your supervisor for more information. These decisions may require the help of a safety professional or someone with chemical safety knowledge.

The meaning of "not available" or "not applicable" on an SDS

With the exception of sections 12 to 15, the supplier is required to provide information on each specific information element required on the SDS. In some cases, it may be appropriate for the supplier to

state “not available” or “not applicable” instead of providing the specific information.

- “Not available” means that the information could not be located or does not exist. For example, if the supplier cannot locate any studies that measure the odour threshold, which is reported in Section 9 of the SDS, the supplier would report “not available.”
- “Not applicable” means that the information element is not relevant. For example, if the product is odourless, then the odour threshold would be reported as “Not applicable.”

Note that the supplier should not use the abbreviation “n.a.” or “NA” without defining it, as it could mean “not applicable,” “not available,” or something entirely different.

MSDS headings vs. SDS headings

The following table compares the section headings of “old” sample MSDSs against the headings of SDSs. Note that there was no set format for MSDSs, but the SDS format is standardized.

Table 8. MSDS headings vs. SDS headings

9-section MSDS headings	16-section MSDS headings	16-section WHMIS 2015 SDS headings
1 Product information	1 Chemical product and company identification	1 Identification
2 Hazardous ingredients	2 Composition/Information on ingredients	2 Hazard identification
3 Physical data	3 Hazards identification	3 Composition/Information on ingredients
4 Fire and explosion data	4 First aid measures	4 First-aid measures
5 Reactivity data	5 Fire fighting measures	5 Fire-fighting measures
6 Toxicological properties	6 Accidental release measures	6 Accidental release measures
7 Preventive measures	7 Handling and storage	7 Handling and storage
8 First aid measures	8 Exposure control / Personal protection	8 Exposure controls/ Personal protection
9 Preparation information	9 Physical and chemical properties	9 Physical and chemical properties
	10 Stability and reactivity	10 Stability and reactivity
	11 Toxicological information	11 Toxicological information
	12 Ecological information	12 Ecological information*
	13 Disposal considerations	13 Disposal considerations*
	14 Transport information	14 Transport information*
	15 Regulatory information	15 Regulatory information*
	16 Other information	16 Other information

* Sections 12 to 15 require the headings to be present, but under Canadian regulations, the supplier has the option to not provide information in these sections.

Looking beyond an SDS for more information

An SDS may not necessarily contain all the information you need. A lot of health hazard information, for example, is written in general terms. In addition, SDSs are often written for many different uses or applications of the product, and the handling and safety precautions may not be specific to your workplace.

Your health and safety committee or representative, health and safety specialist, occupational health nurse, family doctor, supervisor, employer, or supplier should be able to help you find more information if needed.

Updating SDSs

SDSs are required to be accurate at the time of sale or import. An SDS needs to be updated when the supplier becomes aware of any "significant new data."

This means that an SDS must be updated when:

- There is new information that changes how the hazardous product is classified, or
- There are changes to the way you should handle or store the product, or to the way you should protect yourself from the hazards of the product

SDSs need to be updated within 90 days of the supplier being aware of the new information. If you purchase a product within this 90-day time period, the supplier must inform you of the significant new data and the date on which it became available in writing.

How to find out if an SDS has been updated

Every SDS must provide a date of last revision in Section 16 (Other information). You will know if an SDS has been updated by checking this date and comparing it to the one on any previous SDS you have.

Responsibilities for SDSs

Suppliers

Suppliers of hazardous products must do the following:

- Provide an accurate and complete SDS for each hazardous product they sell in Canada.
- Provide an SDS in both official languages (English and French).
- Obtain or prepare an SDS on or before importing a hazardous product.
- Ensure that information is not ambiguous and does not conflict with information on the supplier label or in other parts of the SDS. If abbreviations are used, they must be explained in the text.
- Update an SDS within 90 days of significant new data becoming available.

Employers

Employers will be required to make sure that all hazardous products (as defined by the Hazardous Products Regulations) have up-to-date SDSs when they enter the workplace. The SDSs must be readily available at all times to the workers who are exposed or near to the hazardous products, and to the health and safety committee or representative. (It's not acceptable for an SDS to be read to a worker over a telephone or radio.)

Employers may computerize the SDS information as long as:

- All employees have access to and are trained on how to use the computer or device.
- The computers/devices are kept in working order.
- The employer makes a hard copy of the SDS available to the employee or health and safety committee/representative upon request.

As mentioned earlier, in some circumstances an employer may be required to prepare an SDS (for example, when the product is produced and used in the workplace).

Employers who use hazardous products are also responsible for the following:

- Educating and training workers to ensure they understand the information on SDSs and are able to use the information to work safely with and near hazardous products.
- Ensuring that a hazardous product is not used, stored, or handled in a workplace unless all the applicable WHMIS requirements concerning labels, identifiers, SDSs, and worker education and training are complied with. If these requirements have not been met, the employer can temporarily store the hazardous product while actively working toward compliance.
- Ensuring that no SDS is more than three years old. Chemicals are constantly being studied, and new information can affect the health and safety information on an SDS. Even if the ingredients of a product have not changed, other important information may have changed.

When an up-to-date SDS is unavailable

When an SDS is three years old, an employer must obtain an up-to-date SDS from the supplier.

As an employer, you have the following options if an up-to-date SDS is unavailable:

- Get written confirmation from the supplier that the SDS hasn't changed, or that the most recent SDS doesn't apply to the hazardous product.
- Update the existing SDS by adding any significant new data or information about the hazardous product that you are aware of, based on the product's ingredients.

For more information, refer to section 5.14 of the Occupational Health and Safety Regulation.

Confidential business information ("trade secrets")

Confidential business information (CBI) refers to specific product information that suppliers or employers who are manufacturers are permitted to withhold from an SDS or label for a period of three years. In the United States, CBI may be called trade secrets or proprietary information. Under WHMIS, a supplier can make a request to Health Canada to protect certain information that gives a company an economic advantage over competitors. Crucial information such as health hazards may **never** be withheld.

Types of information that may be granted this exemption are:

- Chemical identity
- Concentration of ingredients
- Information that can be used to identify an ingredient, such as a toxicological study
- Information that can be used to identify the hazardous product
- Information that can be used to identify the supplier of the hazardous product

The acronym HMIRA stands for the federal *Hazardous Materials Information Review Act.*

If the term "trade secret" or "proprietary information" appears on an SDS, it must have an HMIRA registration number and date. The following are examples of statements you may see:

- While a CBI claim is being processed:
  CBI under review. HMIRA Registration No.: 1938. Filing date: July 20, 2018.
- After a CBI claim has been granted:
  HMIRA Registration No.: 1938. Date granted: August 20, 2018.

When information is needed for emergency or first aid treatment, a supplier or employer must immediately disclose — in confidence — to a treating physician or nurse, the specific chemical identity and other necessary information about a hazardous product protected by a trade secret claim. In addition, an officer of Health Canada may disclose — in confidence — withheld information to agencies responsible for occupational health and safety.

For more information on confidential business information and WHMIS, consult the *Hazardous Materials Information Review Act* and Regulations, or contact Health Canada. For links and contact information, see pages 92 and 93.